

customised rulesets for firewalls, LVS (Linux Virtual Server) for load balancing and Heartbeat for HA needs.

Whatever solution you choose, the firewall will be set to allow incoming port 80/tcp connections (and port 443/tcp if you are serving some secure content through SSL), and redirect them to the reverse proxy servers on the DMZ, which are running nginx.

The external firewall on data-centre-1 will load balance the traffic between the reverse-proxy server on each data centre. The algorithm for load-balancing should be wisely chosen. LVS supports round-robin, least connections, weighted round-robin and weighted least connections. Avoid round-robin here for two reasons: one, so that there are more chances of connections being distributed evenly and two, we are already using round-robin load-balancing at both DNS and reverse-proxy levels.

A connection that reached data-centre-1 may either go to reverse-proxy servers located at data-centre-1 or at data-centre-2, depending on the decision taken by the load-balancing algorithm.

Layer 2: nginx reverse proxy servers (DMZ)

The packet redirected by the external firewalls may reach either of the reverse-proxy set-ups at each data-centre. Both the reverse-proxy set-ups are configured identically and contain all the six nodes (Web-servers) as their members. So, irrespective of which reverse-proxy the connection reaches, it may actually be processed by any of the six nodes.

In the first article in this series, we explored the set-up of the nginx reverse-proxy, so I will not be going into too much detail here.

Layer 3: Web servers (internal zone)

The Web servers may run either nginx or Apache, depending on the requirement. Again, we have already covered the configuration of nginx as a Web server with PHP support in the first article itself, so details are not required.

One of the important points here is the website being served. Since quite a lot of the code will be the same, should we copy the same code across all Web servers while deploying the code or should we make use of a network-based common storage pool like NFS or SAN? The answer to this is based on your requirement and whether or not you would like to add a few more blocks in the architecture. For a website, static or dynamic, I would prefer to write a deployment script that can deploy the code to all the Web servers and not use SAN or NFS.

The second important point is the database. If you are serving a dynamic website that uses databases, then you should ideally configure the Web servers located in each data-centre to connect to the database situated locally. This will save the slow SQL queries to be executed over a

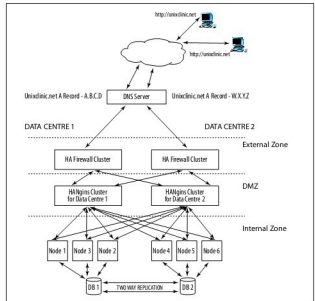


Figure 1: The architecture

WAN link between data centres. The application utilising the database should typically use a connection pool including the database at both the data centres, with a higher priority for the database situated locally.

Layer 4: Databases (internal zone)

The database layer in our architecture is purely optional, but if you do have one, then there are certain points you need to keep in mind. The database queries should only be using the database located in the same data centre where possible, and only in case of its failure should they go to the other data-centre.

Typically, for this kind of set-up, there will be one database in each data centre with a two-way replication set-up between them.

Moving further

In this article, we have seen how we can achieve data centre redundancy to increase the availability of your website.

For the geek in you, I have given enough pointers to get you started. I sincerely hope that you have enjoyed these three articles and will take advantage of them to put this architecture into production, to demonstrate the benefits of using FLOSS to your organisation.

This is definitely not the last of the articles in this series, as we are yet to cover the firewall configuration and data centre load balancing using LVS. We also need to look at setting up a database cluster with two-way replication. This will be covered in forthcoming issues. 

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